

Complete Summary

Project Title

Netflix Movies Data Analysis Using Python

Project Overview

This project analyzes a Netflix movie dataset to discover trends in movie genres, popularity, ratings, and release years. The analysis includes data cleaning, preprocessing, exploratory data analysis (EDA), and visualization using Python libraries such as **Pandas**, **NumPy**, **Matplotlib**, and **Seaborn**.

Objective

The main objectives of this project are to:

- Understand the structure of the Netflix movie dataset.
 - Clean and preprocess the data for analysis.
 - Analyze movie genres and popularity.
 - Identify the highest and lowest popular movies.
 - Categorize movie ratings for better interpretation.
 - Discover movie release trends over the years.
 - Visualize important insights using graphs.
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Dataset Information

- **Rows:** 9,827
- **Columns:** 29

The dataset contains information such as:

- Movie Title
- Genre
- Release Date
- Popularity
- Vote Average

- Vote Count
 - Overview
 - Original Language
 - Poster URL
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Libraries Used

- NumPy
 - Pandas
 - Matplotlib
 - Seaborn
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Data Exploration

The dataset was explored to understand its structure.

The following operations were performed:

- Displayed the first five rows using `head()`.
- Examined column names and data types using `info()`.
- Generated descriptive statistics using `describe()`.
- Checked for duplicate records.
- Verified missing values.

Findings

- The dataset contains **9,827 rows** and **29 columns**.
 - No duplicate records were found.
 - No missing values were present.
 - The dataset was clean and ready for preprocessing.
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Data Preprocessing

Several preprocessing steps were applied to improve the dataset quality.

1. Release Date Conversion

The **Release_Date** column was converted into datetime format, and only the release year was extracted.

2. Dropping Unnecessary Columns

The following columns were removed because they were not useful for analysis:

- Overview
- Original Language
- Poster URL

3. Categorizing Vote Average

The **Vote_Average** column was grouped into four categories:

- Not Popular
- Below Average
- Average
- Popular

This made it easier to compare movies based on audience ratings.

4. Genre Processing

The **Genre** column contained multiple genres separated by commas.

The preprocessing steps included:

- Splitting genres into a list.
- Exploding the DataFrame so that each movie has one genre per row.
- Converting the Genre column into a categorical data type.

Data Visualization

Several visualizations were created to answer important business questions.

1. Most Frequent Genre

A count plot was created to identify the most common movie genre.

Finding:

Drama is the most frequent genre in the dataset.

2. Vote Average Distribution

A count plot was used to analyze the distribution of movie ratings.

Finding:

Most movies fall into the **Popular** category after categorization.

3. Highest Popular Movie

The movie with the maximum popularity score was identified.

Finding:

Spider-Man: No Way Home has the highest popularity.

Genres:

- Action
 - Adventure
 - Science Fiction
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4. Lowest Popular Movie

The movie with the minimum popularity score was identified.

Finding:

The movie with the lowest popularity belongs to genres such as:

- Music
 - Drama
 - War
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5. Movie Releases by Year

A histogram was plotted to analyze movie releases over time.

Finding:

The year **2020** has the highest number of movie releases in the dataset.

Key Insights

- Drama is the dominant genre.
 - Most movies are categorized as **Popular** based on vote averages.
 - **Spider-Man: No Way Home** is the most popular movie.
 - Movies released in **2020** are the highest in number.
 - Splitting genres into separate rows provides more accurate genre analysis.
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Conclusion

This project successfully demonstrates the complete data analysis workflow, including data cleaning, preprocessing, visualization, and insight generation. The analysis reveals that **Drama** is the most common genre, while **Spider-Man: No Way Home** has the highest popularity score. Categorizing movie ratings made it easier to interpret audience preferences, and analyzing release years showed that **2020** had the highest number of movie releases. Overall, this project showcases practical skills in **Python, Pandas, NumPy, Matplotlib, Seaborn, data preprocessing, exploratory data analysis (EDA), and data visualization**, making it an excellent portfolio project for aspiring data analysts.